A Comparative ERGM Analysis of Cold War Alliances (1965–1985)

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Why This Research Matters

- Cold War security structures were deeply shaped by formal alliances.
- Traditional IR models overlook relational and structural dimensions of alliances.
- ERGMs allow modeling tie probabilities in full networks, not just dyads.
- We investigate how nuclear status, military power, and bloc affiliation influence alliance ties.

Research Question

Main Question

To what extent do power-related attributes—nuclear capability, military expenditure, and bloc affiliation—influence the formation of military alliances in 1960?

- Framed within **neorealist theory**: alliances serve state survival and balance of power.
- Modeled using **Exponential Random Graph Models (ERGMs)** to estimate tie probabilities.

Data Sources

- **Formal Alliances Dataset (Correlates of War):** All defense pacts as undirected edges between states.
  https://correlatesofwar.org/data-sets/formal-alliances/
- **National Material Capabilities Dataset (Correlates of War):** Military expenditure as a continuous node attribute.
  https://correlatesofwar.org/data-sets/national-material-capabilities/
- **Manual Coding:**
  - *Nuclear Status:* NWS vs NNWS.
  - *Bloc Affiliation:* NATO, Warsaw Pact, Non-Aligned, Other.

Research Design and Methodology

- Network constructed from 1965-1975-1985 mutual defense pacts.
- **Node attributes:**
  - *nuclear:* Nuclear status" coded as binary (NWS/NNWS); tested via nodecov("nuclear")
  - *milex (military expenditure) :* "Military expenditure" (milex) taken from NMC dataset, logged where needed; entered via nodecov("milex")
  - *bloc:* "Bloc affiliation" categorized into NATO/Warsaw/Non-Aligned/Other; tested via nodefactor("bloc") and nodematch("bloc") (for homophily).
- **ERGM**: Probabilistic modeling of alliance tie formation based on nodal covariates and structural terms.

Research Design and Methodology

Why 1965, 1975, and 1985?

- Captures key phases of the Cold War:
  - **1965**: Vietnam War escalation, peak U.S.-Soviet polarity.
  - **1975**: Post-Vietnam, rise of non-aligned movement.
  - **1985**: Gorbachev's reforms, Soviet bloc fragmentation begins.
- These snapshots allow exploration of structural shifts in alliance patterns.

Why Not Time-Series?

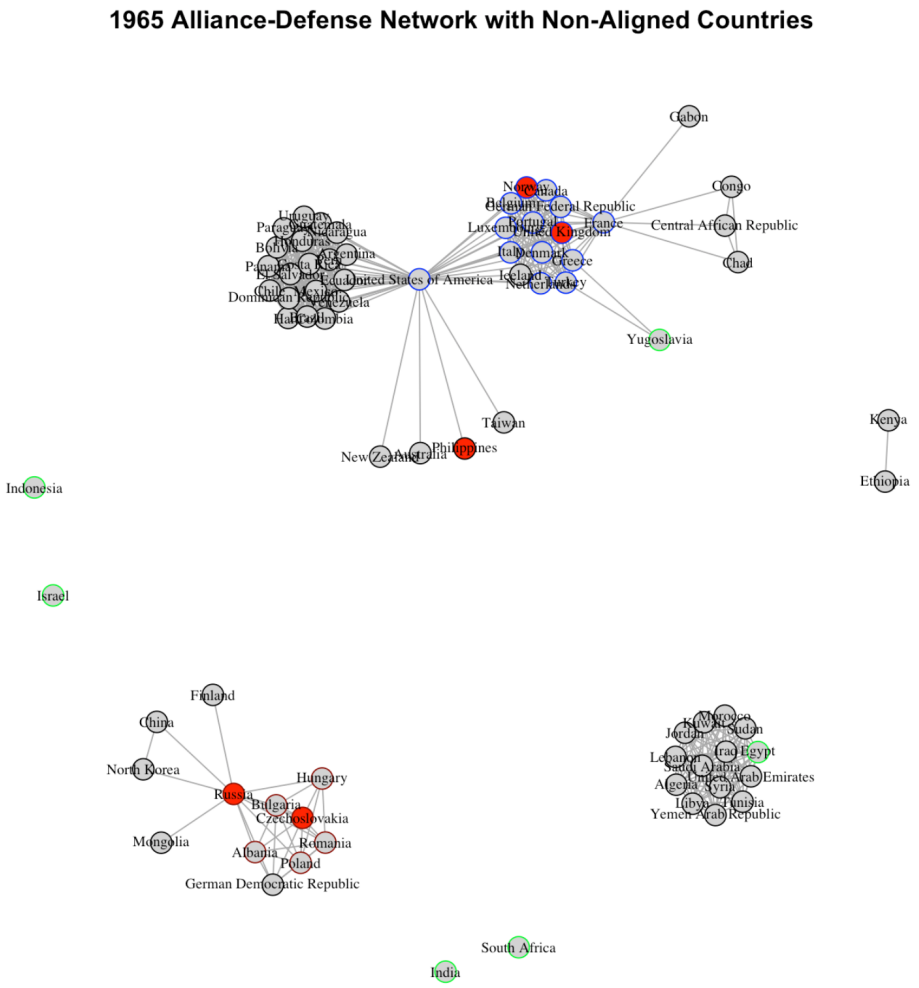
- ERGMs are designed for cross-sectional network analysis.
- Time-series ERGM models (TERGM/STERGM) require advanced methods beyond the scope.
- Limited time and data constraints make dynamic modeling unfeasible.

Research Design and Methodology

Methodological Validity

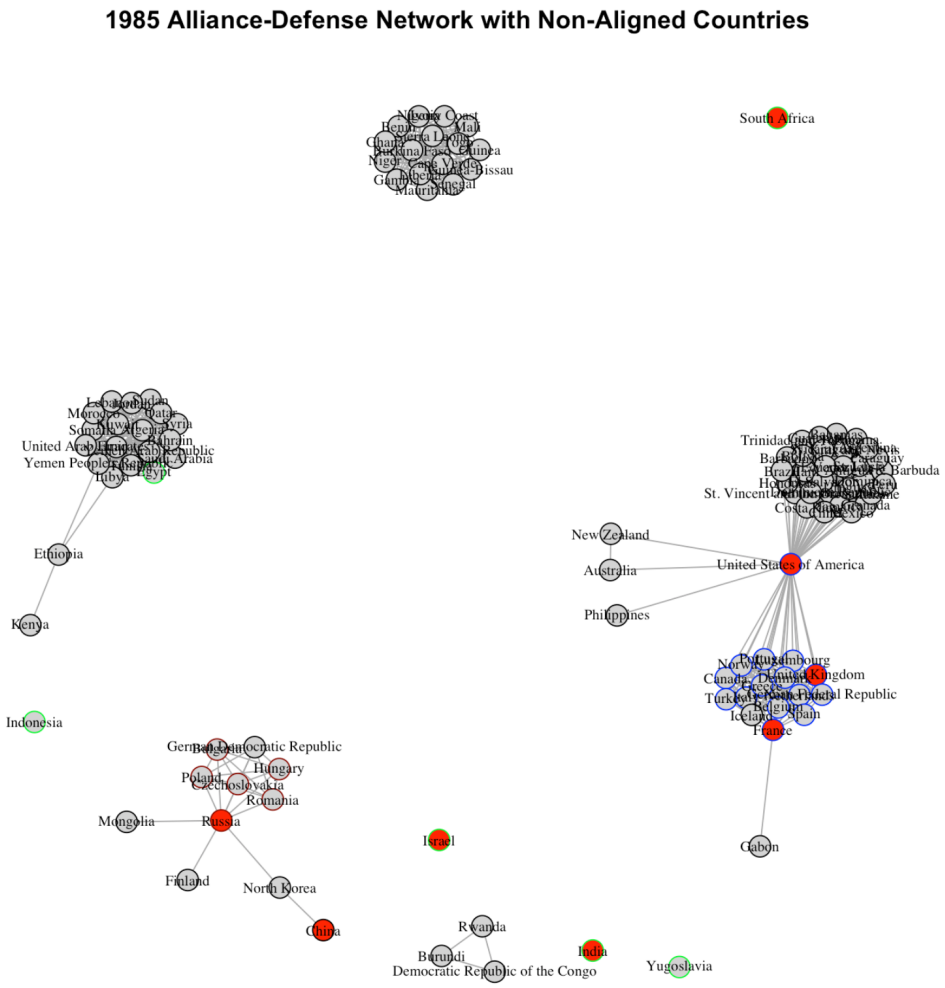
- Each year (1965, 1975, and 1985) is treated as an independent network.
- Allows tracking of coefficient changes and structural metrics over time.
- Facilitates robust comparison of alliance drivers (in this research, nuclear status and milex).

1965 Alliance Network Plot



1975 Alliance Network Plot

1985 Alliance Network Plot



Hypotheses

H1: Nuclear Status Hypothesis

- NWS are more likely to form alliances than NNWS.
- Rationale: Nuclear powers are powerful and attractive allies.
- *Tested via:* `nodecov("nuclear")`

H2: Military Expenditure Hypothesis

- States with higher military expenditure are more likely to form alliances.
- Rationale: Stronger states are more capable of alliance formation.
- *Tested via:* `nodecov("milex")`

ERGM for 1985 Context

Source on Save Run Source

```r
model85 <- ergm(net85_clean ~ edges +
                  nodecov("nuclear") +     # H1: NWS vs. NNWS
                  nodecov("milex") +       # H2: military spending
                  nodefactor("bloc"))      # Control for bloc membership

summary(model85)

#RESULTS
#nodecov.nuclear: estimate = -0.608, p = 0.004

#This result is statistically significant and negative.

#It means: nuclear states were less likely to form alliances in 1985

#Possible interpretation:
#Nuclear states (e.g., US, USSR, China) may rely on self-defense capabilities or bilateral deterrence instead of multilateral alliances.
#OR
#lack of sufficient external securities and systemic insecurity against other powers, states develop nuclear weapons for deterrrence

#H1 is rejected: Nuclear states formed fewer alliances, not more.


#nodecov.milex: estimate = +6.88e-09, p < 0.0001

#Highly significant and positive.

#States with higher military spending were more likely to form alliances.

#The small coefficient is due to large scale of milex (millions), but the direction and significance are robust.

#H2 is supported: Greater military power is associated with more alliance ties.
```

ERGM Results for 1965 (model65)

- **H1 Result:**
  - Coefficient for `nodecov.nuclear` estimate= -0.4947, p = 0.035
  - Statistically significant and *negative*
  - **Interpretation:** Results show NWS forms fewer alliances
- **H2 Result:**
  - Coefficient for `nodecov milex` estimate= 3.425e-08, $p < 0.0001$
  - Positive and statistically significant
  - **Interpretation:** Military expenditure increases the likelihood of alliance formation.

ERGM Results for 1975 (model75)

- **H1 Result:**
  - Coefficient for `nodecov.nuclear` estimate= -0.455, p = 0.0399
  - Statistically significant and *negative*
  - **Interpretation:** Results show NWS forms fewer alliances rather than more
- **H2 Result:**
  - Coefficient for `nodecov milex` estimate= +1.52e-08, $p < 0.0001$
  - Positive and statistically significant
  - **Interpretation:** Military expenditure increases the likelihood of alliance formation.

ERGM Results for 1985 (model85)

- **H1 Result:**
  - Coefficient for `nodecov.nuclear` estimate= -0.608, p = 0.004
  - Statistically significant and *negative*
  - **Interpretation:** Results show NWS forms fewer alliances rather than more
- **H2 Result:**
  - Coefficient for `nodecov milex` estimate= +6.88e-09, $p < 0.0001$
  - Positive and statistically significant
  - **Interpretation:** Military expenditure increases the likelihood of alliance formation.

Conclusion

- All three of the Exponential Random Graph Models support H2 while rejecting H1.
- **Military power** (expenditure) increases alliance likelihood.
- **Nuclear states** form fewer ties rather than more as put forward by H1.
- One possible interpretation would be that nuclear states (e.g., US, USSR, China) may rely on self-defense capabilities or bilateral deterrence instead of multilateral alliances or lack of sufficient external securities and systemic insecurity against other powers, states develop nuclear weapons for deterrence.